

Requirements Table

Problem Statement #12 — Blood Bank Inventory & Emergency Donor Matcher | Actors: Emergency Requester, Blood Bank Manager, Donor, SMS Gateway

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall cross-reference emergency blood unit requests against real-time blood bank inventory and broadcast SMS alerts to compatible donors within a 10 km radius.	High	Pass: Compatible donors notified within 30 seconds of emergency flag. Fail: An incompatible blood group is alerted.	Core emergency-response function; rapid, blood-group-accurate donor mobilization directly drives survival outcomes.
FR-002	Functional	The system shall allow the Blood Bank Manager to register and update blood bag inventory records, capturing blood group, component type, collection date, and expiry date.	High	Pass: A newly registered blood bag appears in the inventory list with a correctly computed expiry date within 5 seconds of submission. Fail: The record is not saved or the expiry date is miscalculated.	Accurate, real-time inventory data is the foundation every other requirement (matching, shortage detection, reporting) depends on.
FR-003	Functional	The system shall automatically flag blood units nearing expiry (within 48 hours) as 'Near-Expiry' and remove expired units from allocatable stock.	High	Pass: Units within 48 hours of expiry are tagged 'Near-Expiry' on the dashboard, and units past their expiry timestamp are removed from allocatable stock within 1 minute. Fail: An expired unit remains allocatable.	Prevents transfusion of unsafe blood and gives the manager lead time to prioritize near-expiry stock for use or disposal.
FR-004	Functional	The system shall allow an Emergency Requester to submit an emergency blood request specifying blood group, component type, quantity, and hospital location.	High	Pass: The submitted request is logged with a unique Request ID and timestamp, and is visible to the Blood Bank Manager within 5 seconds. Fail: The request is not logged, or is accepted with a required field missing.	A structured, complete request is required before the system can search inventory or match donors.
FR-005	Functional	The system shall allow an eligible donor to confirm or decline availability by replying to an emergency SMS alert, updating the donor's status in the system in real time.	Medium	Pass: A donor reply ('YES'/'NO') updates the donor's availability status within 60 seconds, visible to the Blood Bank Manager. Fail: A donor reply is not reflected in the system.	Confirms which donors are actually coming, letting the manager avoid duplicate dispatch calls and focus follow-up on unconfirmed donors.
NFR-001	Nonfunctional (Performance & Security)	The inventory ledger must maintain strict transactional consistency, preventing simultaneous allocation of the same blood bag.	High	Pass: Benchmarking tests confirm target latency and security standards are met under simulated peak load, with no double-allocation of a single blood bag. Fail: Two concurrent requests are able to reserve the same blood bag.	Double-allocation during a shortage could mean a bag promised to two patients — a direct patient-safety and data-integrity risk.
NFR-002	Nonfunctional (Reliability / Availability)	The system shall be available at least 99.9% of the time, measured over a rolling 30-day window.	High	Pass: Uptime monitoring logs show availability of at least 99.9% over any 30-day window ($\leq \sim 43$ minutes downtime/month). Fail: Downtime exceeds 43 minutes in a 30-day window.	Emergency blood requests can occur at any hour; extended downtime could directly delay a life-saving transfusion.

Use Case: UC-01 — Request Emergency Blood Unit

Primary Actor: Emergency Requester

Supporting Actors/Systems: Blood Bank Manager, Donor, SMS Gateway

Includes: UC-02 (Search Compatible Donors) → UC-03 (Send SMS Alert to Donors)

Extends: UC-06 (Verify Donor Eligibility) optionally extends UC-03

Preconditions

- The Emergency Requester is authenticated in the system (hospital/requester account).
- The blood bank inventory database and donor registry are operational and reachable.
- At least one donor record with a valid phone number and blood group exists in the donor registry.

Postconditions

- Success: A blood unit is reserved from stock, or one or more compatible donors have been alerted and the request status is 'Notified'/'Fulfilled', and the request is logged with a Request ID.
- Failure: The request is logged with status 'Cancelled' or 'Escalated', and the Blood Bank Manager has been notified.

Main Success Scenario

1. The Emergency Requester selects "Request Emergency Blood Unit" from the portal/app.
2. The system displays a request form for blood group, component type, quantity, and hospital location.
3. The Requester enters the required details and confirms the hospital location (auto-detected via GPS or entered manually).
4. The system validates the inputs (blood group format, quantity > 0, location present).
5. The system creates an emergency request record with a unique Request ID, timestamp, and priority flag "Critical".
6. The system «include»s UC-02 Search Compatible Donors: it queries current inventory and the donor registry for the requested blood group within a 10 km radius of the hospital location.
7. If sufficient inventory stock exists, the system reserves the matching blood unit(s) and notifies the Blood Bank Manager to prepare dispatch; the flow continues at step 9.
8. If stock is insufficient, the system «include»s UC-03 Send SMS Alert to Donors, broadcasting an SMS to all eligible, compatible donors within the 10 km radius via the SMS Gateway.
9. The system displays a confirmation to the Requester containing the Request ID and an estimated response window.
10. Use case ends successfully once the required units are reserved/dispatched or donor confirmations are logged against the request.

Alternate Flow — 4a. Invalid Blood Group Entered

- 1. The system detects that the entered blood group is not a recognized type (e.g., not one of A+, A-, B+, B-, AB+, AB-, O+, O-).
- 2. The system displays an "Invalid Blood Group" error and highlights the field.
- 3. The Requester corrects the entry; the flow resumes at step 3.

Alternate Flow — 9a. No Donor Response Within SLA

- 1. If no donor confirms availability within 15 minutes of the SMS alert, the system automatically widens the search radius to 20 km and re-triggers UC-03 Send SMS Alert to Donors.
- 2. The system notifies the Blood Bank Manager that the request has been escalated.
- 3. The flow resumes at step 8 with the wider radius; if still unresolved after one further SLA window, the request status is set to